Sadman Sakib

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About Me

Industry-experienced software engineer and M.Sc. student in Computer Science and Engineering with research experience in human-centered AI, computer vision, and assistive technology, and a passion for bridging theory and practice. My research focuses on designing intelligent systems, drawing on HCI, multimodal AI, and NLP, that support people whose needs are often overlooked by mainstream technology. A collaborative team player with a strong problem-solving mindset, sharpened through competitive programming (Codeforces Specialist, 1100+ problems solved), who enjoys helping others learn.

Research Interests

Human-Computer Interaction · Accessibility · Human-Centered AI & Design · Computer Vision · Long Video Understanding · NLP & Large Language Models · Multimodal AI

Education

M.Sc. in Computer Science and Engineering 2026 – Present
BRAC University, Dhaka, Bangladesh *CGPA: 4.00/4.00 (12 credits completed)*

B.Sc. in Computer Science and Engineering 2019 – 2024
Khulna University of Engineering & Technology (KUET), Bangladesh *CGPA: 3.73/4.00*

Technical Skills

Languages: Python, C, C++, Java
ML & Vision: PyTorch, Hugging Face, OpenCV, NumPy, Pandas
Backend: Spring Boot, Kafka, RabbitMQ
Databases: PostgreSQL, MySQL, Oracle, Redis
Tools: Git, LaTeX, Jira, Confluence
AI Tools: Claude, Cursor

Publications

An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading
13th International Conference on Electrical and Computer Engineering (ICECE), 2024 [IEEE Xplore]

- Designed and developed a portable embedded smart spectacle system to assist visually impaired individuals with mobility and reading support.
- Implemented real-time obstacle avoidance, fall detection with automated caregiver alerts, and an OCR-based text recognition pipeline using image preprocessing and text-to-speech conversion.

Mobile Application to Collect Data and Measure Blood Component Level in a Non-Invasive Way
26th International Conference on Computer and Information Technology (ICCIT), 2023 [IEEE Xplore] [GitHub]

- Developed a smartphone-based mobile application for non-invasive estimation of blood glucose and hemoglobin levels using photoplethysmography (PPG) signal analysis.
- Implemented real-time video acquisition and integrated an existing deep learning model to extract and post-process PPG signals for non-invasive estimation of glucose and hemoglobin levels.

Automated Object Identification for the Visually Impaired
27th International Conference on Computer and Information Technology (ICCIT), 2024 [IEEE Xplore]

- Developed a unified assistive system for visually impaired individuals, integrating object detection, currency recognition, and face recognition into a single automated platform.

- Trained and implemented deep learning models (YOLOv5 for detection and ResNet-50 for classification) using curated datasets to enable real-time environmental awareness and identity recognition.
- Designed modular pipelines for object localization, currency categorization, and facial recognition to support navigation, financial transactions, and social interaction.

Current Research

AI-Assisted Video Creation for People Who Stutter *Manuscript under review*

- Conducted a qualitative HCI study with eight Bangladeshi people who stutter, combining an onboarding workshop with semi-structured interviews and reflexive thematic analysis, to explore how they create videos and use AI in the video creation journey.
- Studied how the purpose of a video, its audience, and personal preference influenced whether participants kept or edited out their stuttering.
- Found that participants want AI tools that lower editing effort while letting creators stay in control of how their voice is represented.

Dataset Contributions

Object Recognition Dataset: a curated image dataset for object recognition, publicly released on Kaggle. [Kaggle]

Industry Experience

Graduate Engineer, Kona Software Lab Ltd. Aug 2024 – Aug 2026
Fintech: backend engineering for national-scale payment systems *Dhaka, Bangladesh*
Nagad: Mobile Financial Service

- Worked on the microservice backend of one of Bangladesh's leading mobile financial services, building customer-facing and admin-portal features.
- Delivered transaction processing, bulk disbursement, and Mastercard Payment Gateway Services (MPGS), NPSB, and Bangla QR integrations.
- Worked with the client team and cross-functional stakeholders on requirements analysis, system design, and performance testing.
- *Stack:* Java, Spring Boot, JPA, Hibernate, JUnit, Oracle, Redis, Kafka, RabbitMQ.

Academic Projects

Query-Guided Grid Composition for Long Video Understanding (Python, PyTorch, VLMs) [GitHub]
Training-free, query-aware frame composition method that improves long-video understanding and question answering in VLMs.

Vikings Chess (Hnefatafl) (Python, Pygame) [GitHub]
Turn-based two-player board game with an AI opponent that uses minimax with alpha-beta pruning.

Online Quiz Management (Java Swing, MySQL) [GitHub]
Desktop app where students take MCQ and written exams and teachers set questions and submit marks.

Sweet Home (HTML, CSS, JavaScript, PHP) [GitHub]
E-commerce site for household furniture with cookie-based recommendations and an admin panel.

Honors & Activities

- **Dean's Award** for excellent academic performance, KUET: 2nd, 3rd, and 4th year.
- **Competitive Programming:** Codeforces Specialist, max rating 1442 (sadman15019); solved 1100+ problems across Codeforces, VJudge, AtCoder, CodeChef, LeetCode, and LightOJ.
- **Workshop Manager**, Special Group Interested in Programming Contest (SGIPC), KUET 2022 – 2023